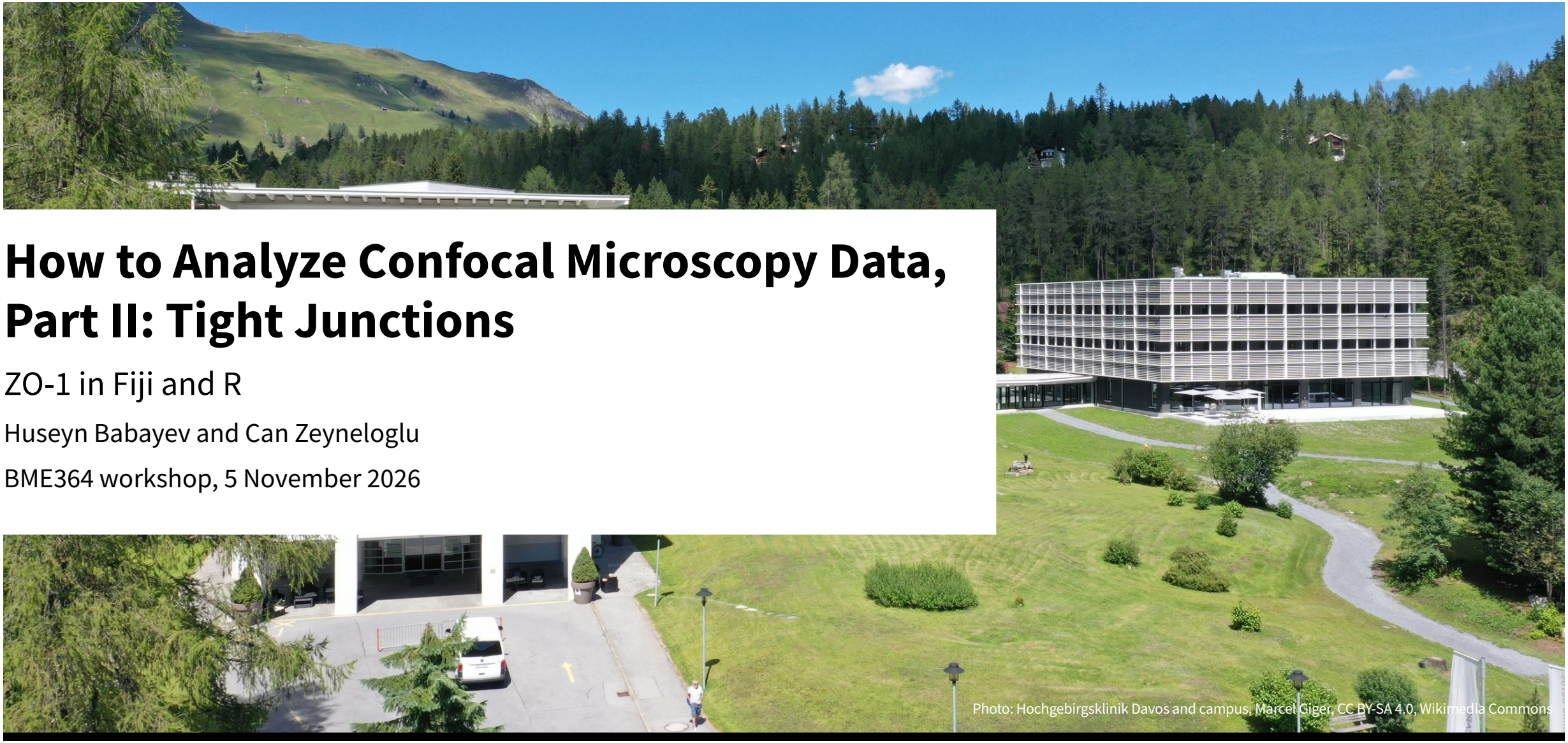
Universität Zürich[UZH]

Swiss Institute of Allergy and Asthma Research (SIAF)



How to Analyze Confocal Microscopy Data, Part II: Tight Junctions

ZO-1 in Fiji and R

Huseyn Babayev and Can Zeyneloglu

BME364 workshop, 5 November 2026

Photo: Hochgebirgsklinik Davos and campus, Marcel Giger, CC BY-SA 4.0, Wikimedia Commons

Teaching plan

4 Nov: Part I

Fiji basics with JC-1

- Pixels, bit depth, integrity
- Threshold → watershed → Analyze Particles
- Red : green ratio per cell

Today: Part II

Tight junctions with ZO-1

- Z-stack → projection
- ZO-1 and DAPI mean per field
- Fields → wells → statistics in R

Davos: Group I

Your own experiment

- Caco-2: P80 0, 0.02, 0.1, 0.5 %
- ZO-1 + DAPI, 24-well glass-bottom plate
- 5 wells × 5 random fields, confocal

Course materials

github.com/drbabayev/UZH-BME364-G1-share

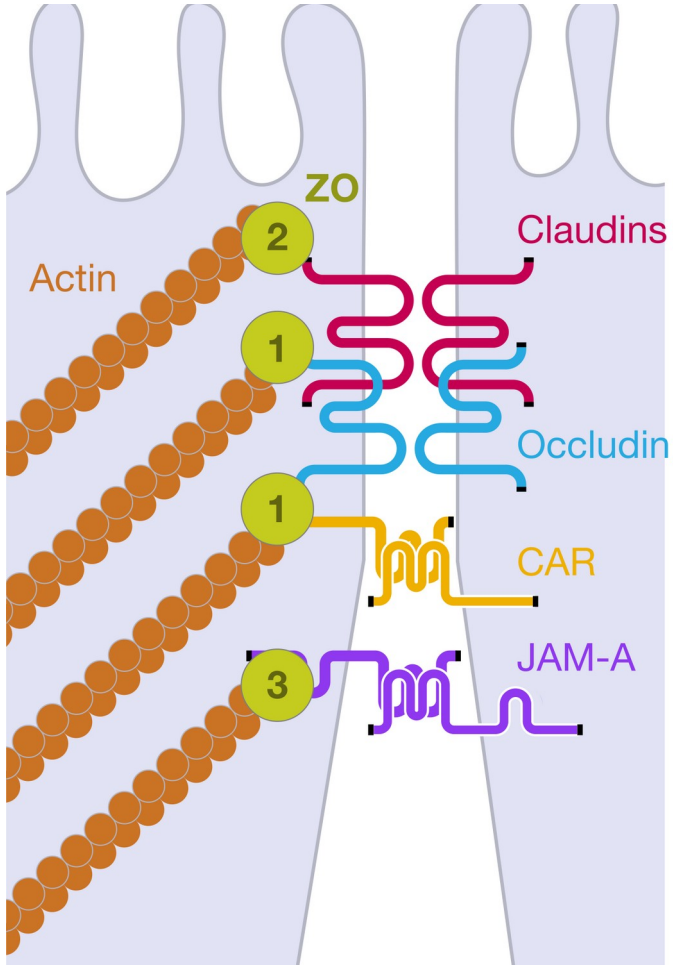
Measuring ZO-1 in Fiji

Theory, data structure, algorithm, implementation

01

ZO-1 is the scaffold of every tight junction

Theory | Data | Algorithm | Implementation



Zeyneloglu, Babayev et al., FEBS Lett 2025, Fig. 2B;
doi:10.1002/1873-3468.70113

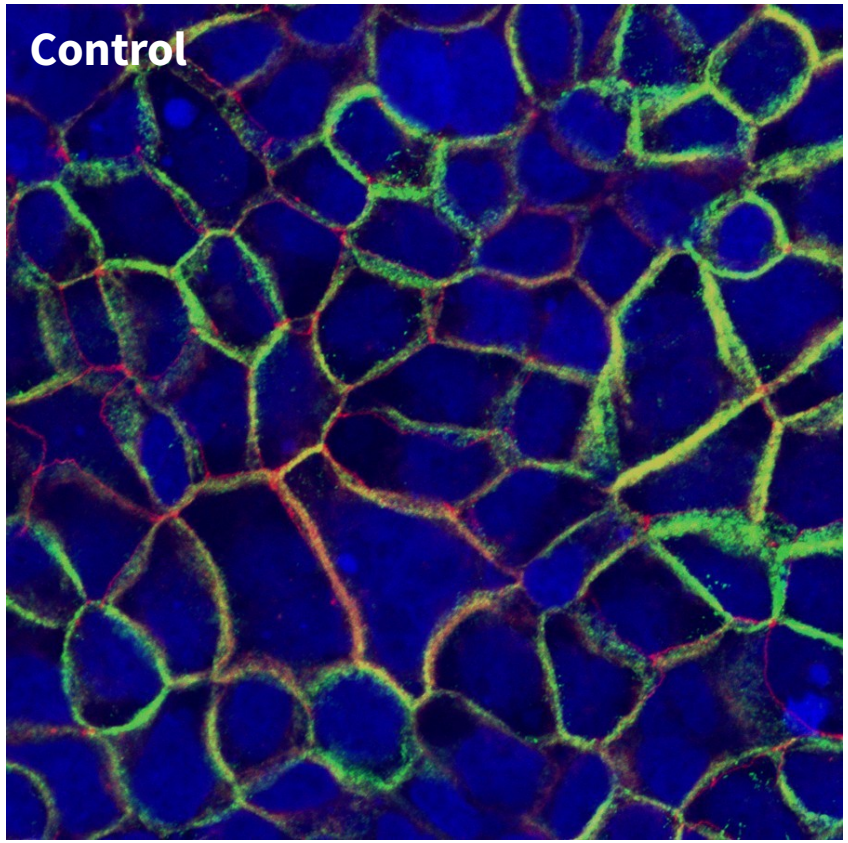
ZO-1 is the scaffold

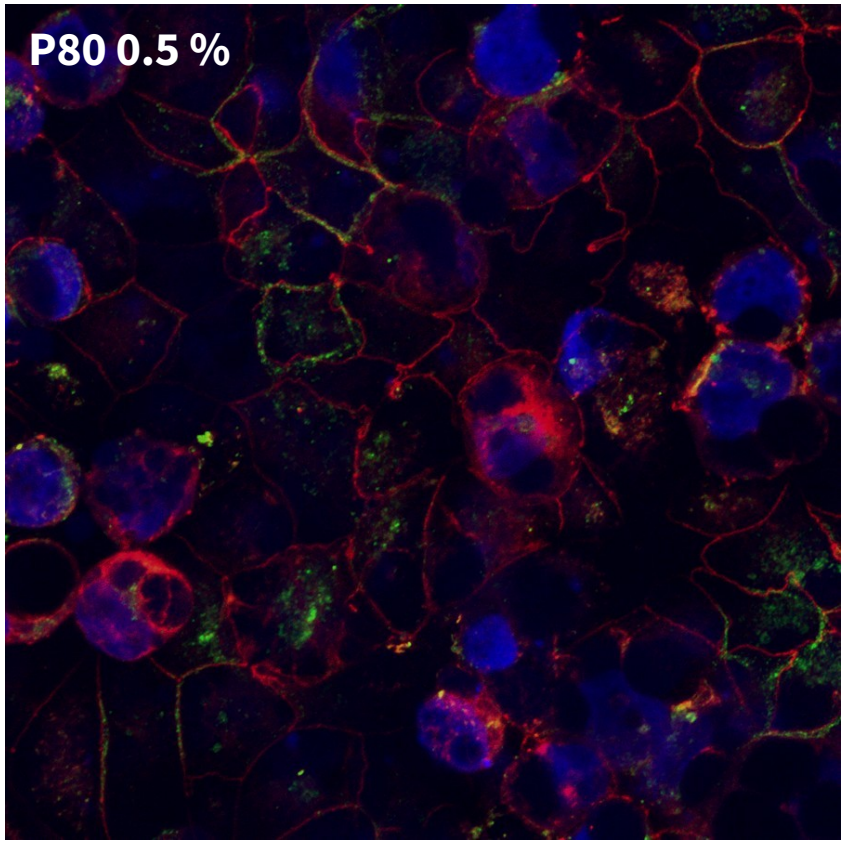
ZO-1, ZO-2 and ZO-3 bind claudins, occludin, CAR and JAM-A and link them to actin: ZO-1 outlines every cell as a continuous honeycomb.

P80 opens the barrier

Polysorbate 80 is a detergent: the junctions loosen, ZO-1 falls and the honeycomb breaks up. At high doses the cells die.

What P80 does to a real monolayer





Real fields, earlier experiment: Caco-2, occludin + claudin-1, DAPI blue; max projection

Control

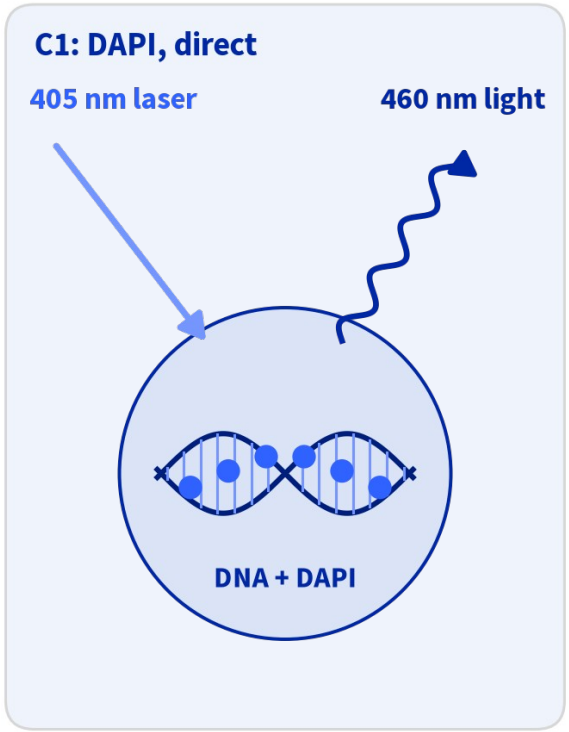
Closed honeycomb: every cell outlined, junctions continuous.

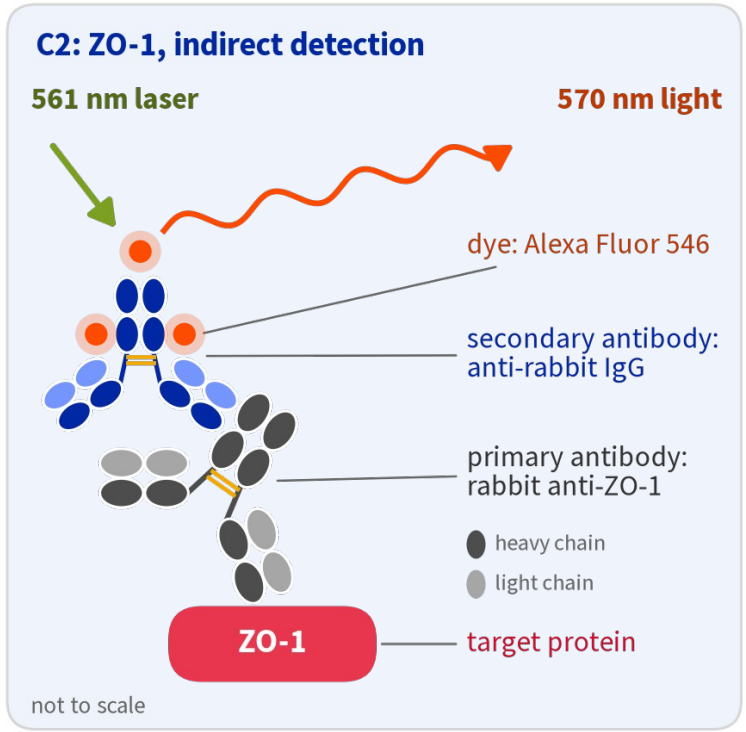
P80 0.5 %

Network broken, weaker junction signal, cells round up: MTT shows no living cells.

Immunofluorescence: from ZO-1 to a signal

1 Fix	2 Permeabilise	3 Block	4 Primary	5 Secondary	6 DAPI
4 % PFA, 15 min	0.1 % Triton X-100, 10 min	3 % BSA, 1 h	rabbit anti-ZO-1, 1:100	anti-rabbit Alexa 546, 1:500	1 µg/mL, nuclei





Indirect detection

Several labelled secondaries bind each primary: a brighter signal.

Light in, light out

One laser per dye, longer light out: C1 DAPI (405 nm), C2 ZO-1 (561 nm).

Confocal pinhole

Blocks out-of-focus light: thin optical sections; a z-stack covers the junctions.

Same settings

Same laser, gain, pinhole and zoom for every well; no saturated pixels (HiLo LUT).

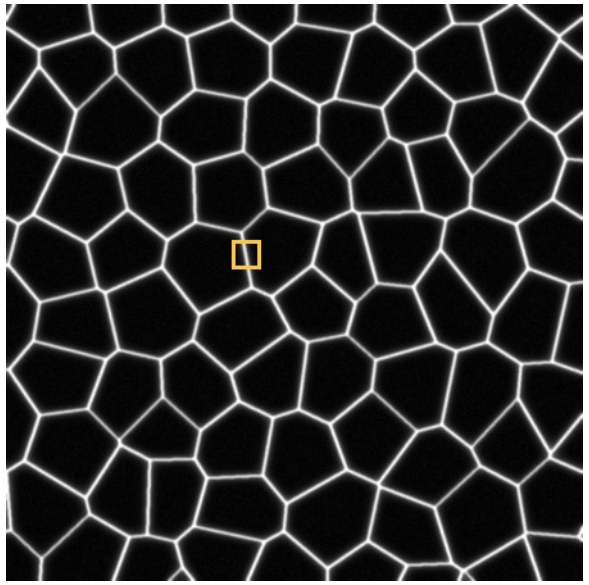
No-primary background

Secondary only, same settings: its mean is subtracted from every field.

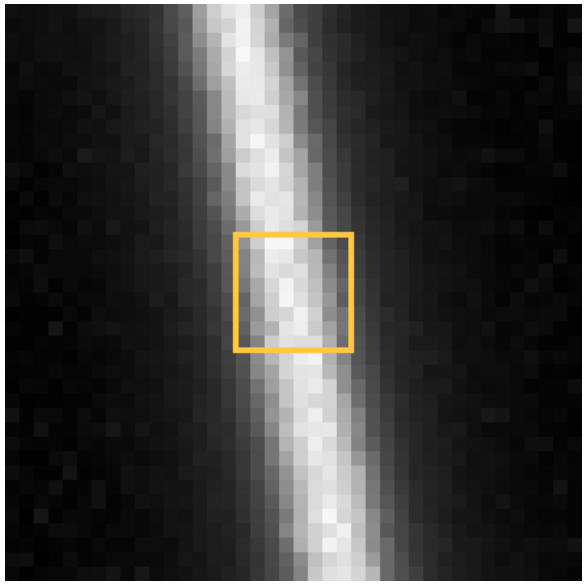
Two means per field

DAPI and ZO-1, whole field, raw. **Read-out:** the ZO-1/DAPI ratio.

An image is a grid of numbers



ZO-1 channel, 900 × 900 px = 90 × 90 µm



40 × 40 pixels (4 × 4 µm): each square is one pixel

1726	2352	2776	2774	2387	1881	1256	991
1581	2123	2666	2624	2294	2116	1391	1071
1555	2163	2576	2481	2541	2129	1565	1146
1454	1968	2448	2729	2472	2200	1670	1274
1252	1923	2366	2758	2535	2250	1778	1184
1327	1720	2356	2438	2709	2373	1690	1414
1206	1828	2068	2600	2673	2312	1898	1437
1118	1574	2340	2555	2523	2518	2035	1336

8 × 8 pixel values = what Fiji measures

Junction

high values (≈ 2700) in the middle columns

Cytoplasm

low values (≈ 1000) at the edges

Fiji

the status bar shows the value under the cursor

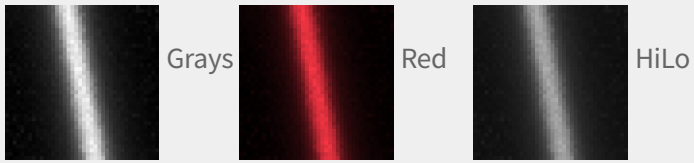
Pixel size = calibration

0.1 µm per pixel, stored in the file metadata (Image › Properties). Area = pixel count × 0.01 µm².

Bit depth = value range

8-bit 0–255, 12-bit 0–4095, 16-bit 0–65 535. A pixel at the maximum is saturated: its true value is lost.

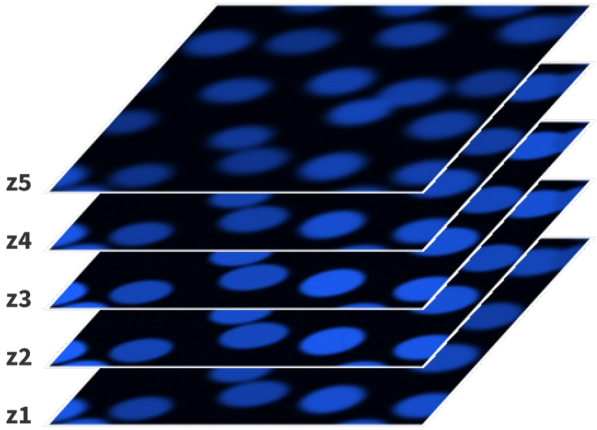
LUT = colors, not data



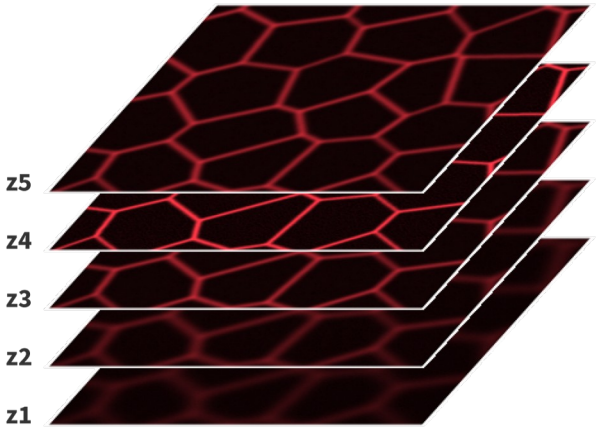
same numbers, HiLo: red = saturated, blue = 0

A confocal file is a hyperstack: X, Y, C, Z

C1: DAPI (nuclei)



C2: ZO-1 (junctions)



5 z-slices, 0.5 µm apart, nuclei sharp at z2–z3, ZO-1 at z4 (apical)

Image › Show Info: the file's dimensions

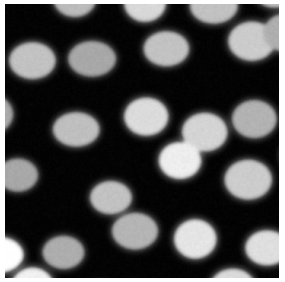
Dimension	Size	Calibration
X (width)	900 px	0.1 µm/px → 90 µm
Y (height)	900 px	0.1 µm/px → 90 µm
C (channels)	2	DAPI, ZO-1
Z (slices)	5	0.5 µm steps
T (frames)	1	one time point
Bit depth	16-bit	2 bytes per value

900 × 900 × 2 C × 5 Z × 2 bytes = 16.2 MB

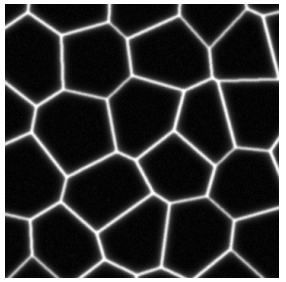
Max projection: per pixel, the highest value along z

z1	z2	z3	z4	z5	→	MAX
593	891	1582	**2776**	1576	→	**2776**

One ZO-1 pixel on a junction: in focus at z4. Use the same slices for every field.

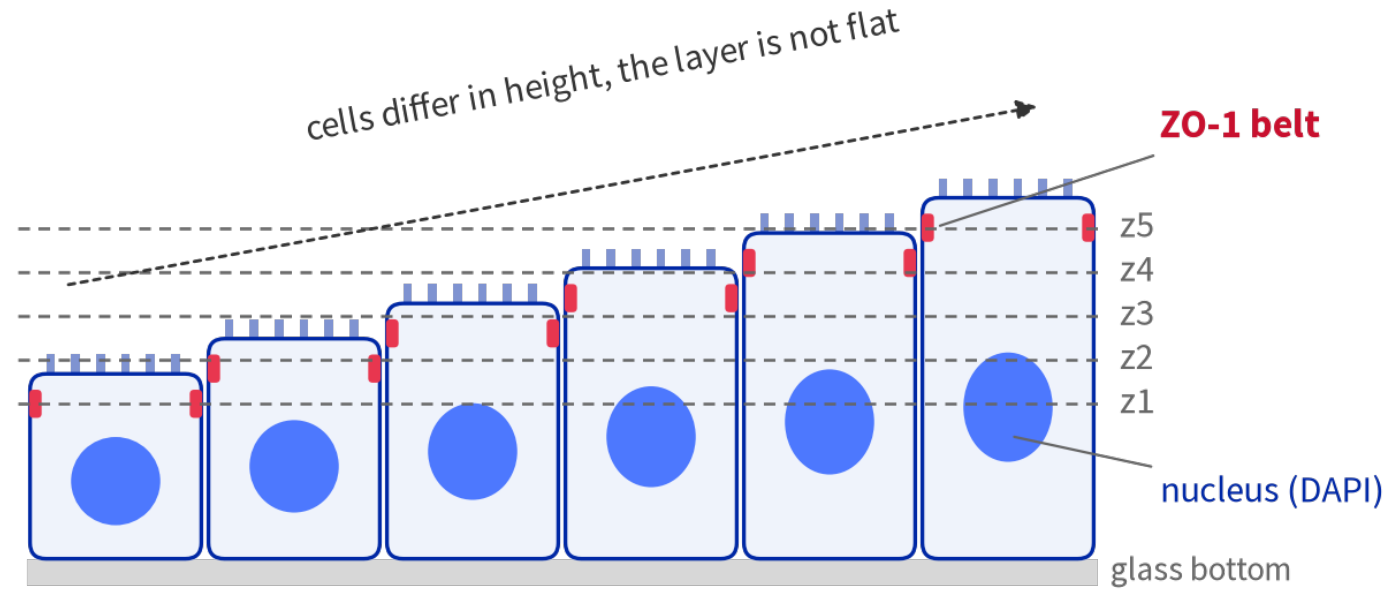


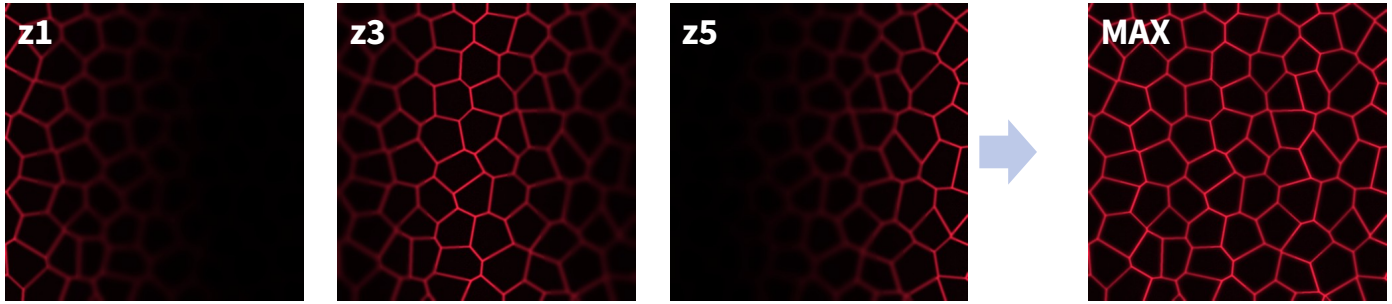
C1-MAX: DAPI



C2-MAX: ZO-1

Why max projection: junctions sit at different heights





Simulated ZO-1, tilted control field: one slice is sharp in one band; MAX shows every junction

Junctions at different heights

Cells differ in height and the layer is not flat: the ZO-1 belt lies at z1 in one cell and at z5 in another.

One slice misses junctions

Out of focus, junctions blur and fade: an intact barrier would look broken.

Max intensity projection

Per pixel, the brightest value along z: every junction at its best focus. Same slices for every field.

From plate to Results table

Condition: control, P80 0.02, 0.1, 0.5 %

Well: 5 per condition → this is n

Field: 5 random, non-overlapping

Z-stack → max projection = one image = one row

Fiji will give us results.csv: measurements, one row per image

Label	DAPI_Mean	ZO1_Mean
zo1_001.tif	512.9	788.6
zo1_051.tif	455.3	656.5
zo1_101.tif	515.3	90.6

You write image_metadata.csv: what each image is

file	plate_well	condition	p80_pct	well	field	primary_ab
zo1_001.tif	A1	ctrl	0	well1	1	ZO-1
zo1_051.tif	C1	P80-0.1	0.1	well1	1	ZO-1
zo1_101.tif	A6	noprimary	0	well1	1	none

- **Metadata table describes the data:** one row per image, filled in while imaging
- **File name = key only:** it links the two tables and carries no design
- **Tidy format:** one observation per row, one variable per column

Per field: project, split, measure

1 Project

`Image › Stacks › Z Project… Max Intensity`

Same slices for every field

2 Split channels

`Image › Color › Split Channels`

C1 = DAPI, C2 = ZO-1

3 DAPI mean

`C1: Analyze › Measure`

Whole field, raw

4 ZO-1 mean

`C2: Analyze › Measure`

Whole field, raw

One row per field → results.csv: Label, DAPI_Mean, ZO1_Mean

In R: ratio = (ZO1_Mean − no-primary ZO-1) ÷ DAPI_Mean, then the mean of the 5 fields of each well

Avoid: measuring after auto-contrast or “Enhance Contrast”, and “Mean / Area × 1000”: Mean is already per pixel; dividing by area again just penalises large regions.

ZO-1 amount: the ZO-1/DAPI ratio

$$\text{ZO-1/DAPI} = (\text{ZO-1 mean}_{\text{field}} - \text{ZO-1 mean}_{\text{no-primary}}) \,/\, \text{DAPI mean}_{\text{field}}$$

We use: ZO-1/DAPI ratio

- ZO-1 per nuclear DNA ≈ ZO-1 per cell
- Gaps lower ZO-1 and DAPI together: they cancel
- Two whole-field means: no segmentation, nothing to tune

Alternative

- Segment nuclei as in Part I: Otsu → watershed → Analyze Particles
- Grow them by ~8 µm = area covered by cells
- ZO-1 mean inside that area; nuclei count as a bonus

Rationale

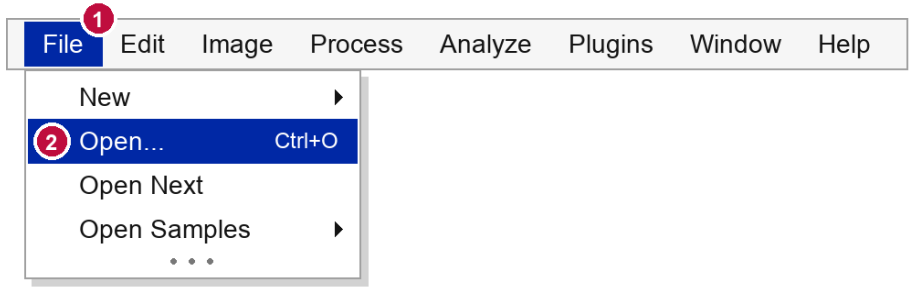
- No watershed or growth radius to check per image
- Fair while cells stay viable (MTT ≥ 94 % up to 0.1 %)
- DAPI raw and never saturated

Assumption: every nucleus gives the same DAPI in all groups. Condensed, apoptotic nuclei are brighter: at 0.5 % P80, where the cells die, read the ratio with care.

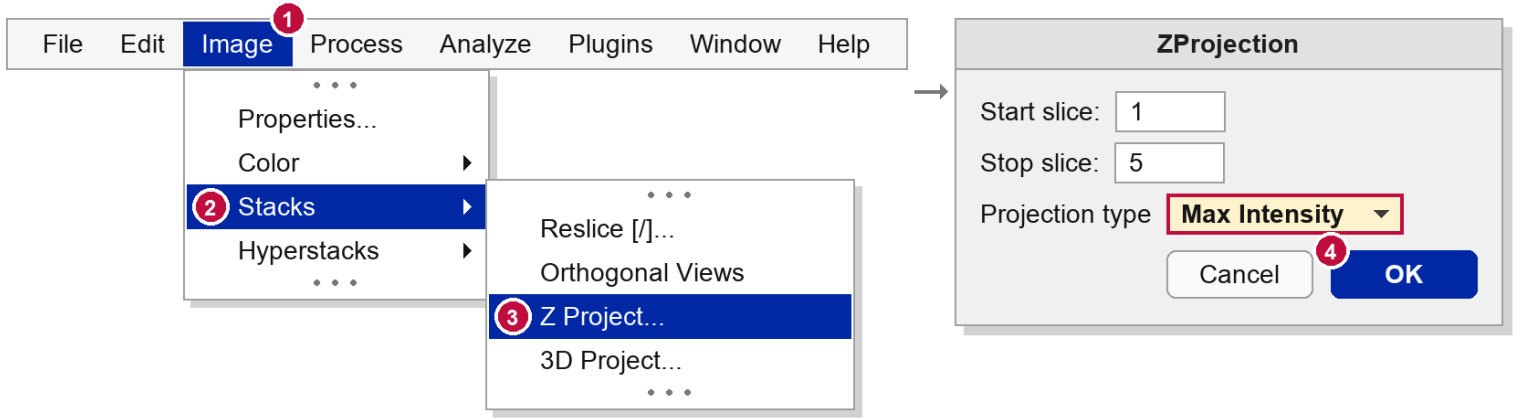
Fiji manually 1: open and project

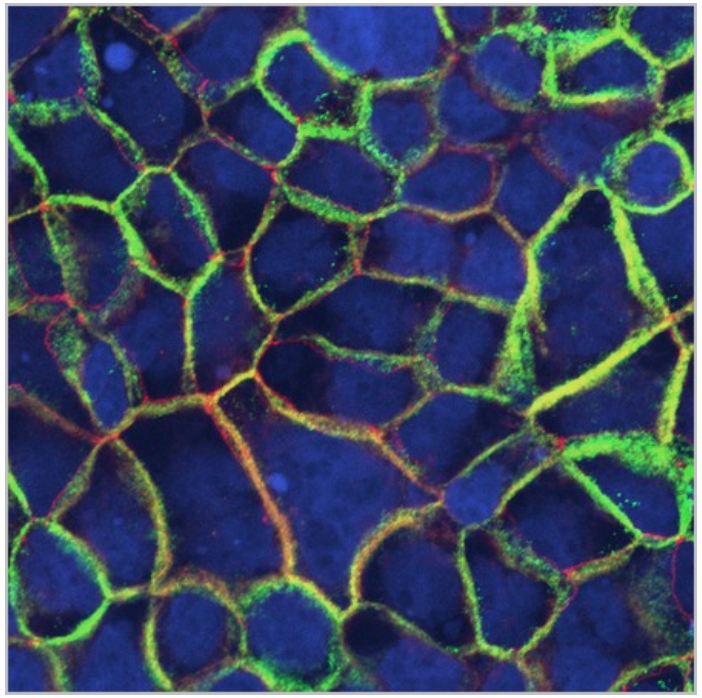
1 Open the TIFF, project the z-stack

Open the .tif (2 channels × 5 slices), or drag it onto Fiji



Max Intensity over all 5 slices





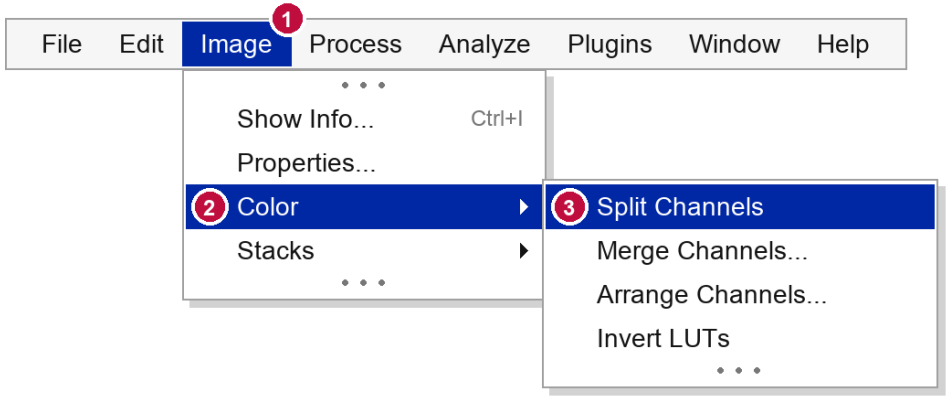
if_001.tif: real field, already projected

Menus drawn after Fiji 2 (on macOS they sit in the menu bar at the top of the screen; Cmd for Ctrl), pictures: example_images/if_001.tif (control, central part), the result of each step

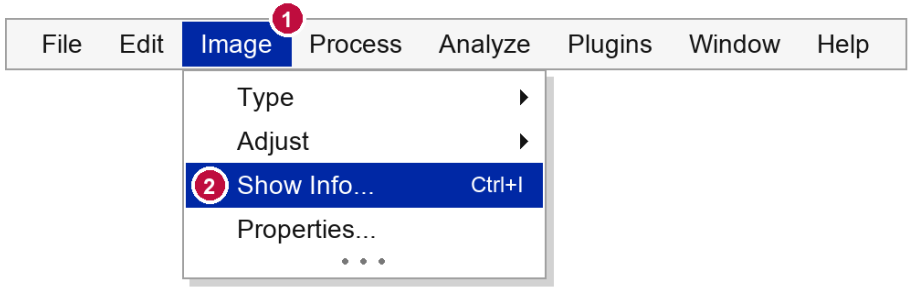
Fiji manually 2: split the channels

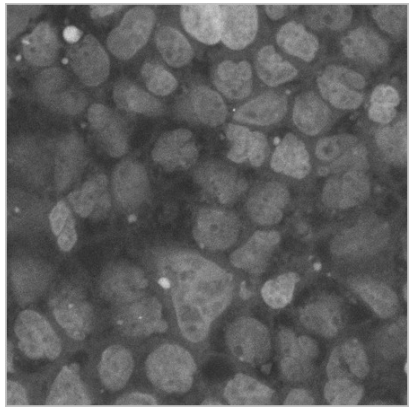
2 Split the channels, check the file

One grayscale image per stain: C1 = DAPI, C2 = ZO-1

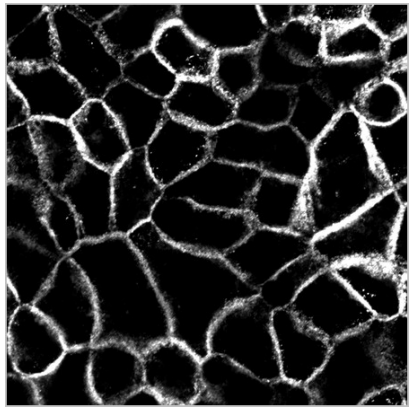


Check: pixel size 0.1 µm, 2 channels, 5 slices





C1: DAPI



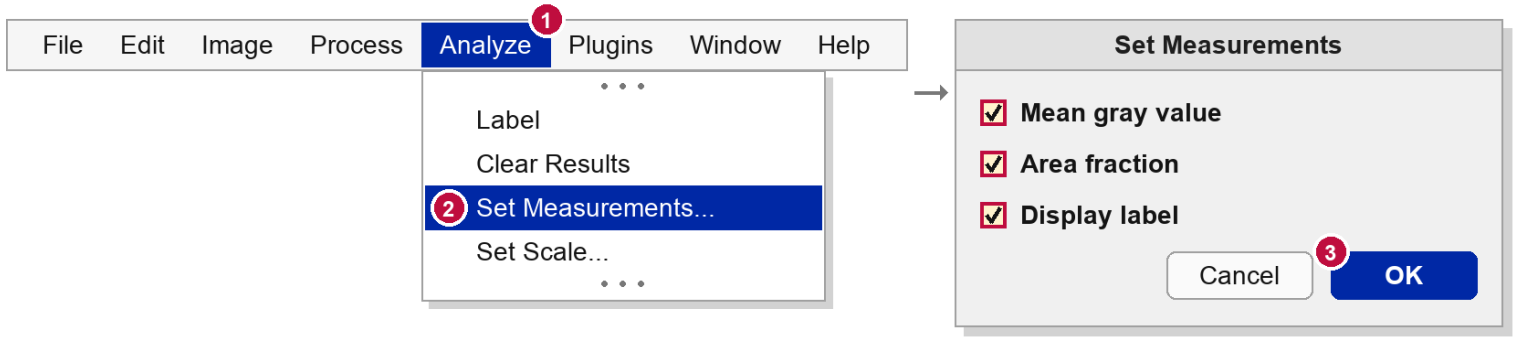
C2: junction stain

Menus drawn after Fiji 2 (on macOS they sit in the menu bar at the top of the screen; Cmd for Ctrl), pictures: example_images/if_001.tif (control, central part), the result of each step

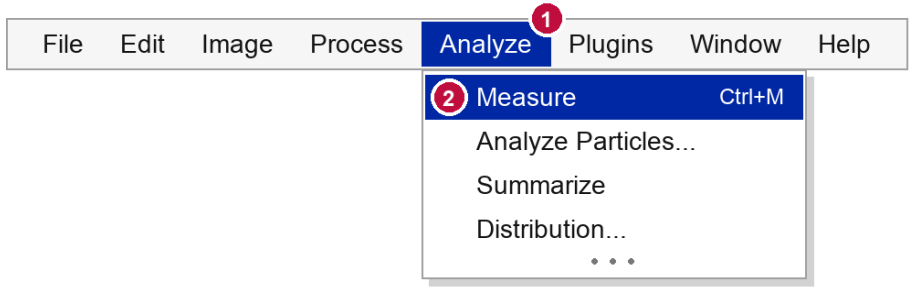
Fiji manually 3: DAPI for the ZO-1/DAPI ratio

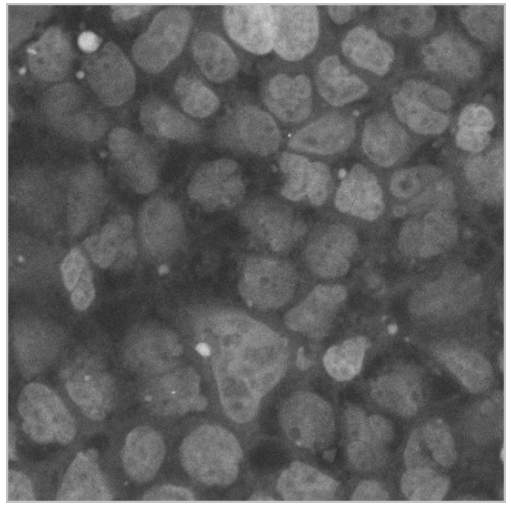
3 Measure DAPI on C1

Once: Mean gray value, Area fraction, Display label



Click C1 (DAPI), nothing selected: Measure, raw values





C1: raw

91.8

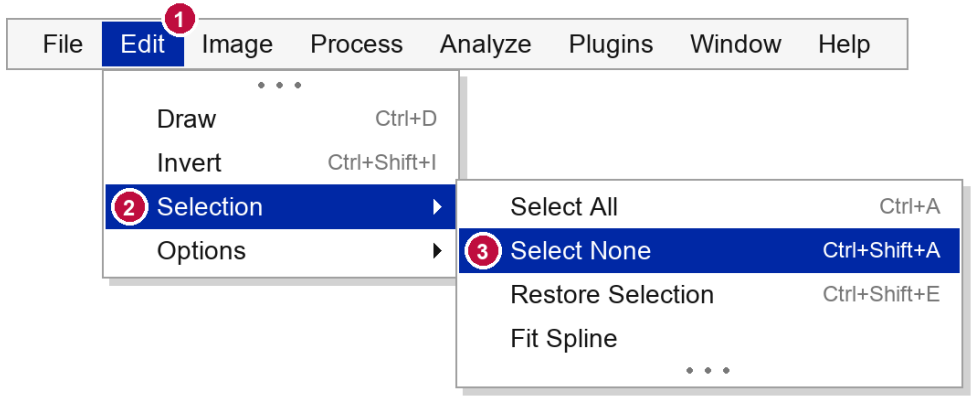
DAPI mean, whole field

Menus drawn after Fiji 2 (on macOS they sit in the menu bar at the top of the screen; Cmd for Ctrl), pictures: example_images/if_001.tif (control, central part), the result of each step

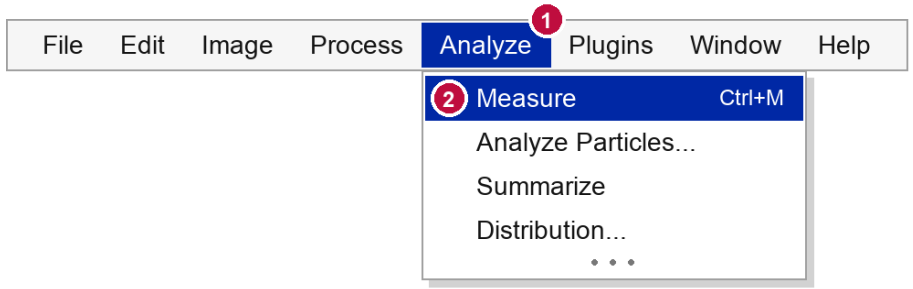
Fiji manually 4: ZO-1 for the ratio

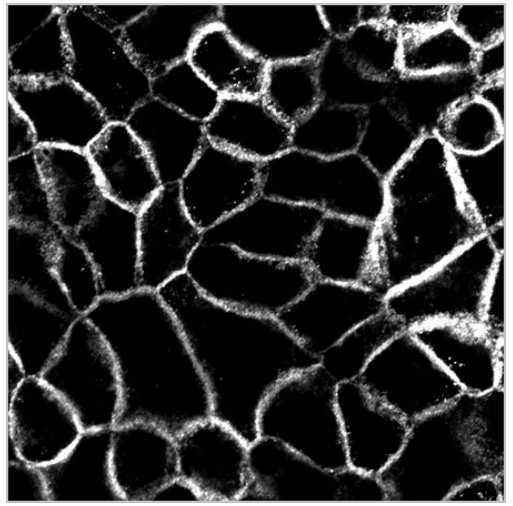
4 Measure raw ZO-1 on C2

Click C2 (ZO-1): make sure nothing is selected



Measure the whole field: raw values





C2: raw

24.3

C2 mean, whole field

R: (ZO-1 − no-primary) ÷ DAPI

Menus drawn after Fiji 2 (on macOS they sit in the menu bar at the top of the screen; Cmd for Ctrl), pictures: example_images/if_001.tif (control, central part), the result of each step

Fiji manually 5: one row per field

5 Same steps on every field

1–2 Project, split the channels

3 C1 (DAPI): Measure, whole field, raw

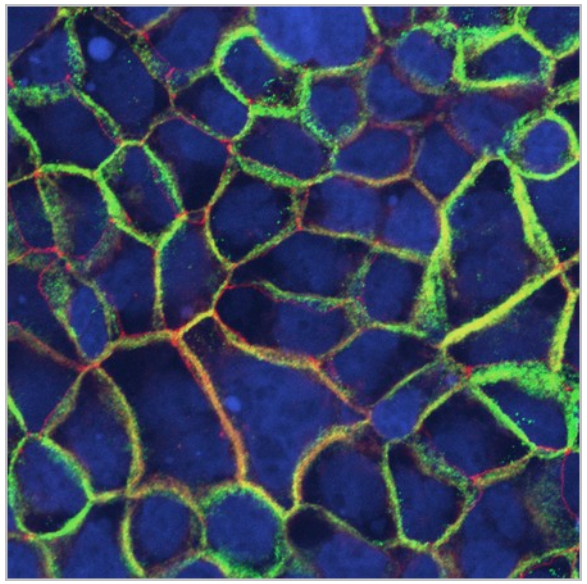
4 C2 (ZO-1): Measure, whole field, raw

Results table: one row per field

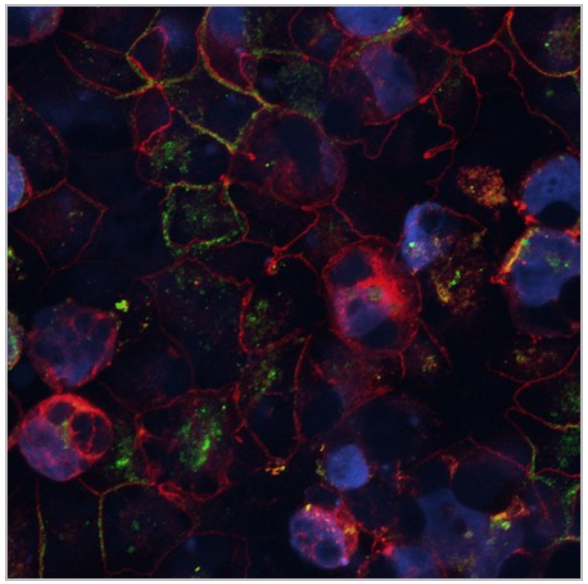
Label	DAPI_Mean	ZO1_Mean
if_001.tif	91.8	24.3
if_004.tif	31.5	3.6

Then in R: ratio = (ZO1_Mean − no-primary) ÷ DAPI_Mean, averaged per well

Control, if_001.tif



P80 0.5 %, if_004.tif



Menus drawn after Fiji 2 (on macOS they sit in the menu bar at the top of the screen; Cmd for Ctrl), pictures: example_images/if_001.tif (control, central part), the result of each step

The Fiji macro

ImageJ macro

```
1  input = getDirectory("Choose the image folder");
   files = getFileList(input);
   for (i = 0; i < files.length; i++) {
     open(input + files[i]);
2    run("Z Project...", "projection=[Max Intensity]");
     img = getTitle();
     run("Split Channels");
3    selectWindow("C1-" + img);              // DAPI
     getStatistics(area, dapi);              // raw DAPI mean
4    selectWindow("C2-" + img);              // ZO-1
     getStatistics(area, zo1);               // raw ZO-1 mean
5    Table.set("Label", row, title, "ZO1");
     Table.set("DAPI_Mean", row, dapi, "ZO1");
     Table.set("ZO1_Mean", row, zo1, "ZO1");
     row++;  run("Close All");
   }
   Table.save(input + "results.csv");
```

1 Loop over the folder

Pick the folder once; the loop opens every image in turn. Every field gets exactly the same steps and parameters: the point of a macro.

2 Project and split

Max intensity over all z-slices → one 2-D image per channel: C1 = DAPI, C2 = ZO-1.

3 DAPI mean

No selection, so the mean covers the whole field. Raw values, no Subtract Background.

4 ZO-1 mean

Whole field, raw values. R subtracts the no-primary value and divides by DAPI.

5 One row per field

Three columns per image; after the loop the table is saved as results.csv.

Run the macro and check its output

results.csv: one row per field

Label	DAPI_Mean	ZO1_Mean
zo1_001.tif	512.9	788.6
zo1_002.tif	503.2	815.1
zo1_076.tif	250.3	318.6
zo1_101.tif	515.3	90.6

Example dataset

4 conditions × 5 wells × 5 fields + 5 no-primary fields = 105 rows, in data/fiji_results_zo1.csv and in data/image_metadata.csv.

Run

- Plugins › Macros › Run… → pick the image folder
- Keep image_metadata.csv complete: one row per file, written while imaging

Check before a batch run

1. **DAPI not saturated**
   Range indicator on C1: the ratio needs real DAPI values
2. **ZO-1 not saturated**
   Brightest control junctions below the maximum value
3. **No-primary imaged the same way**
   Same laser, gain and pinhole: it is the background
4. **Metadata complete**
   One row per image in image_metadata.csv, written while imaging

ZO-1 analysis in R

Theory, data structure, algorithm, implementation

02

Fields are not independent: nested data



4 conditions × 5 wells × 5 fields = 100 fields, but n = 5 wells per condition

Pseudoreplication

Counting 100 fields as n inflates the evidence: fields of one well share seeding, P80 dose, staining and imaging.

Option 1: aggregate

Average fields → wells; compare the 5 well means per condition (ANOVA + Dunnett).

Option 2: model the nesting

Linear mixed model: condition as fixed effect, random intercept for each well.

Show both levels

SuperPlot: all fields in the background, well means on top (Lord et al., J Cell Biol 2020).

Join the metadata to the results

results.csv (Fiji)

Label	DAPI	ZO1
zo1_001.tif	512.9	788.6
zo1_002.tif	503.2	815.1
zo1_051.tif	455.3	656.5

image_metadata.csv (you)

file	plate_well	condition	field
zo1_001.tif	A1	ctrl	1
zo1_002.tif	A1	ctrl	2
zo1_051.tif	C1	P80-0.1	1

join by file name = Label

file	plate_well	condition	DAPI	ZO1	ratio
zo1_001.tif	A1	ctrl	512.9	788.6	1.33
zo1_002.tif	A1	ctrl	503.2	815.1	1.41
zo1_051.tif	C1	P80-0.1	455.3	656.5	1.21

```R
fields <- inner_join(metadata, results,
                     by = join_by(file == Label)) %>%
  mutate(ratio = (ZO1_Mean - bg) / DAPI_Mean)   # bg: no-primary
```

Well means: this is n

condition	plate well	ZO-1/DAPI
ctrl	A1	1.39
ctrl	A2	1.87
ctrl	A3	1.96
…	…	…
P80-0.5	D1	0.95
P80-0.5	D2	0.56
P80-0.5	D3	0.65

Two tables, two levels

- fields: 100 rows → mixed model
- well_means: 20 rows → ANOVA + Dunnett, SuperPlot

Import, subtract, summarise, test, plot

1
Import
results + metadata, join by file

2
Ratio
(ZO-1 − no-primary) ÷ DAPI

3
Summarise
fields → well means

4
Test
well means or mixed model

5
Plot
SuperPlot: fields + wells

Test what was replicated

- ANOVA + Dunnett on the 5 well means per condition (residual df = 16)
- Mixed model on all fields: random intercept per well; Kenward–Roger df = 16

Pseudoreplication: avoid

- ANOVA + Dunnett on 100 fields: df = 96
- The p-value then reflects the number of fields you imaged, not the number of wells

Analysis using R (1/3): from two tables to well means

R

```r
results  <- read_csv("data/fiji_results_zo1.csv") %>%
  select(-any_of(c(" ", "...1")))      # Fiji row numbers
metadata <- read_csv("data/image_metadata.csv")
fields <- inner_join(metadata, results,
                     by = join_by(file == Label))

bg <- fields %>% filter(condition == "noprimary") %>%
  pull(ZO1_Mean) %>% mean()
fields <- fields %>% filter(condition != "noprimary") %>%
  mutate(ratio     = (ZO1_Mean - bg) / DAPI_Mean,
         condition = factor(condition, levels = conds),
         well_id   = plate_well)

well_means <- fields %>%
  group_by(condition, well, well_id) %>%
  summarise(ratio    = mean(ratio),
            n_fields = n(), .groups = "drop")
```

1 Read both tables, join them by file name

read_csv() reads Fiji's results.csv and your metadata table. inner_join() matches every Label to its file row and adds plate well, condition, well and field to each measurement.

2 Background, then the ZO-1/DAPI ratio

bg = mean ZO-1 of the no-primary fields: what secondary antibody and autofluorescence give without ZO-1. For every other field: ratio = (ZO-1 − bg) / DAPI, ZO-1 per nuclear DNA. factor() puts the control first, the reference for Dunnett; well_id (A1 … D5) names each well uniquely.

3 Fields → wells: one value per replicate

group_by() + summarise() average the 5 fields of each well: 100 field rows become 20 well rows, each with its mean ratio. n_fields checks that every well has its 5 fields. These 5 well means per condition are the n.

Analysis using R (2/3): three tests, one right n


```R
# WRONG: 100 fields counted as n, residual df = 96
fit_fields <- aov(ratio ~ condition, data = fields)
emmeans(fit_fields, trt.vs.ctrl ~ condition)$contrasts

fit_wells <- aov(ratio ~ condition, data = well_means)
emmeans(fit_wells, trt.vs.ctrl ~ condition)$contrasts %>%
  summary(infer = TRUE)                    # df = 16

fit <- lmer(ratio ~ condition + (1 | well_id), data = fields)
anova(fit, ddf = "Kenward-Roger")
emmeans(fit, trt.vs.ctrl ~ condition,
        lmer.df = "kenward-roger")$contrasts
```

4 The classic mistake: fields as n

aov() fits a one-way ANOVA; emmeans(trt.vs.ctrl) gives the Dunnett contrasts, each P80 concentration vs control. On fields the model believes it has 96 df, so the SE is too small: P80 0.1 %: p = 0.025.

5 Right: Dunnett on the well means

Same model, one row per well: 20 wells − 4 groups = 16 df. summary(infer = TRUE) adds the 95 % CI of each difference. P80 0.1 %: −0.18, p = 0.34; P80 0.5 %: p < 0.001.

6 Right: mixed model on all fields

lmer() keeps the 100 fields; (1 | well_id) gives every well its own intercept, so fields of one well count together. Kenward–Roger sets the df from the wells (16): the same p-values as step 5, and the well-to-well scatter is estimated.

Analysis using R (3/3): the SuperPlot


```r
p <- ggplot(fields, aes(condition, ratio)) +
  geom_jitter(aes(colour = well), width = 0.15,
              alpha = 0.45, size = 1.8) +
  stat_summary(data = well_means, fun = mean, geom = "crossbar",
               width = 0.5, colour = "#333333") +
  geom_point(data = well_means, aes(fill = well, shape = well),
             size = 4.5, colour = "#333333",
             position = position_dodge(width = 0.35)) +
  scale_colour_manual(values = well_cols) +
  scale_fill_manual(values = well_cols) +
  scale_shape_manual(values = c(21, 22, 24, 23, 25)) +
  scale_x_discrete(labels = labels) +
  labs(x = NULL, y = "ZO-1 / DAPI ratio",
       colour = "Well", fill = "Well", shape = "Well",
       caption = "Simulated data; dots = fields, symbols = wells")
ggsave("figures/05_if_superplot.png", p,
       width = 6.6, height = 4.3, dpi = 200, bg = "white")
```

7 Fields in the background

ggplot() puts condition on x and the ratio on y. geom_jitter() draws all 100 fields, spread sideways and coloured by their well; alpha = 0.45 keeps them faint.

8 Bar = mean of the wells

stat_summary() on well_means, not on fields: the bar is the mean of the 5 well means, the value the test compares.

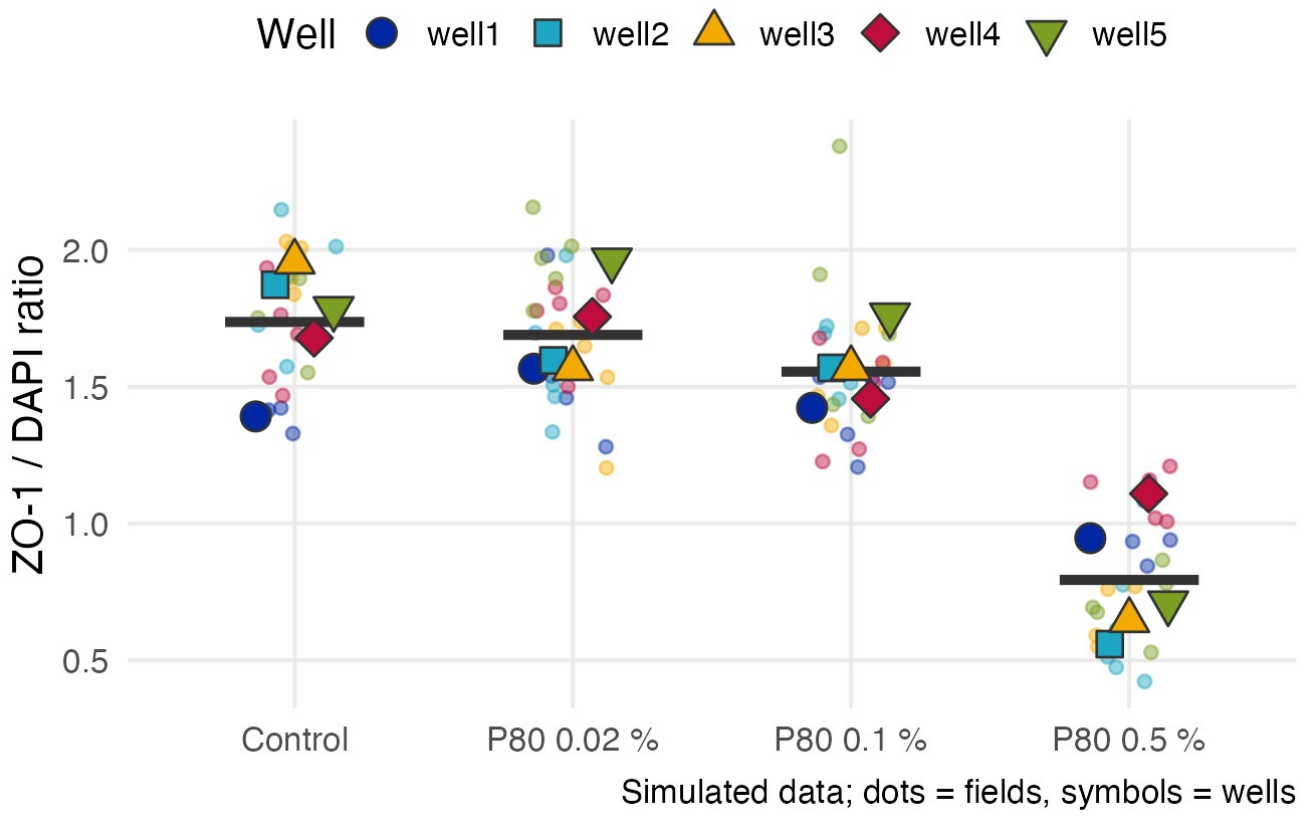
9 Well means on top

One large symbol per well, in the colour of its fields; position_dodge() sets the 5 wells side by side. These are the n.

10 Colours, shapes, labels, save

Each well gets one colour and one shape, so the plot also reads in grey. ggsave() writes figures/05_if_superplot.png.

Result: the SuperPlot



Read it in two layers

Small dots: fields; large symbols: the 5 well means per condition: the n of the test.

ZO-1/DAPI (Dunnett, wells)

Control 1.74; 0.02 % 1.69 ($p = 0.94$); 0.1 % 1.55 ($p = 0.34$); 0.5 % 0.79 ($p < 0.001$).

Not significant ≠ no effect

At 0.1 % the ratio is 10 % lower, but the 95 % CI includes 0, so five wells cannot settle it.

Next to TEER and flux

At 0.1 % TEER falls and flux rises clearly; ZO-1 per cell changes less.

Same fields, two analyses, opposite conclusions

ZO-1/DAPI: P80 0.1 % vs control (Dunnett)	n	Difference	df	p
ANOVA on all fields (pseudoreplication)	100 fields	−0.18	96	0.025
ANOVA on well means	20 wells	−0.18	16	0.34
Mixed model: random intercept per well	100 fields in 20 wells	−0.18 ± 0.12	16	0.34

0.025

what 100 “independent” fields claim

0.34

what 5 wells per condition support

Lesson

Same estimate, opposite conclusion. Counting fields turns a well-to-well scatter into a “significant” effect.

Before you report ZO-1 data: a checklist

1. Identical acquisition settings for every well, including the no-primary control
2. Z-stack projection stated; no saturated pixels in DAPI or ZO-1
3. Fields chosen at random on DAPI, before looking at ZO-1
4. Macro saved with the analysis; metadata table complete
5. Raw intensities, no auto-contrast; no-primary background subtracted
6. n = wells of one experiment (say so); fields averaged or modelled
7. Well means or mixed model, test named in the legend
8. SuperPlot; representative images shown with the same display range

Run everything yourself

```
BME364-G1-share/
├── README.md          start here
├── R/                 mac_linux/, windows/ (.Rmd)
├── data/              example data (.csv)
├── example_images/    4 real IF fields (.tif)
├── fiji/              measure_zo1.ijm
├── figures/           plots, written by R
├── results/           tables, written by R
├── manuals/           lab manuals (.docx)
└── slides/            both decks (.pptx)
```

R

```
# Fiji: Plugins > Macros > Run...
#   fiji/measure_zo1.ijm  -> results.csv

# R: R/mac_linux/ or R/windows/
# once: Source 00_install_packages.R
# open 05_if_analysis.Rmd in RStudio
#   and click Knit
```

Your own data from Davos

- R packages: rmarkdown, tidyverse, rstatix, lme4, lmerTest, emmeans, drc
- Replace the files in data/ with your own exports and keep the column names; the .Rmd files then reproduce every step of this workshop.